

THE OCTANT

Octans



SIZE RANKING 50

BRIGHTEST STAR
Nu (ν) 3.8

GENITIVE Octantis

ABBREVIATION Oct

HIGHEST IN SKY AT 10 PM
OctoberFULLY VISIBLE
0°–90°S

This constellation, which originally was also known as *Octans Nautica* or *Octans Hadleianus*, contains the south celestial pole. It was introduced in the 18th century by the French astronomer Nicolas Louis de Lacaille (see p.422).

The area of sky in which Octans lies is quite barren. Within naked-eye range, the nearest star to the south celestial pole is Sigma

(σ) Octantis. It is of only magnitude 5.4 and thus far from prominent.

Because of the effect of precession (see p.64), the positions of the celestial poles are constantly changing. As a result, the south celestial pole is moving farther away from Sigma and toward the constellation of Chamaeleon. There are no bright stars in this area either, so the region of the south celestial pole will remain blank for another 1,500 years, when the pole will pass just over a degree away from 4th-magnitude Delta (δ) Chamaeleontis.

Octans represents an instrument known as an octant, which was used by navigators to help them find their position (see panel, below). It was invented by the English instrument maker John Hadley (1682–1744).

SPECIFIC FEATURES

Lambda (λ) Octantis is a double star that is divisible with a small telescope. The components are of 5th and 7th magnitudes.



SOUTHERN STAR TRAILS

Curving star trails, drawn out by the Earth's rotation on this long-exposure photograph, emphasize the barren nature of the area around the south celestial pole.

AT THE POLE

Octans comprises only a scattering of faint stars. There is no bright star to mark the southern pole, which lies near the center of this image.

EXPLORING SPACE

NAVIGATION

In 1731, the British mathematician John Hadley built a device called a doubly reflecting octant. The navigator sighted the horizon through a telescope and adjusted a movable arm until the reflected image of the Sun or a star overlay the direct view of the horizon. The altitude of the Sun or star could be read off a scale, from which the navigator could deduce his latitude.



OCTANT

This wood and brass octant is by Browning of Boston. In later designs, the arc was extended from one-eighth of a circle to one-sixth, and the octant became the modern sextant.

